Linear Algebra & Neural Network Foundations

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1. Linear Transformations in 2D

Stretching (Uniform Scaling):
Matrix: [[a, 0], [0, a]]
Effect: (x, y) → (ax, ay)
Determinant: a^2 (area scales by a^2)

Horizontal Shear:
Matrix: [[1, m], [0, 1]]
Effect: (x, y) → (x + my, y)
Determinant: 1 (area preserved)

Rotation:
Matrix: [[cosθ, -sinθ], [sinθ, cosθ]]
Effect: Rotates vectors counter-clockwise by θ
Determinant: 1 (area preserved, length preserved)

2. Forward Propagation (Single Neuron)

Dataset shape: X (n_x, m)
Weights: W (1, n_x)
Bias: b (1, 1)

Linear model:
Z = W @ X + b

If using regression:
Y_hat = Z

If using logistic regression:
Y_hat = sigmoid(Z)

3. Mean Squared Error (MSE) Cost

Cost function:
$J = (1 / (2m)) * \Sigma(Y\_hat - Y)^2$

Why 1/2?
It simplifies the gradient during differentiation.

Used for:
• Linear regression
• Continuous outputs

4. Training Loop Structure

Each iteration consists of:

1. Forward pass → compute Y_hat
2. Compute cost → measure error
3. Backprop/update → adjust parameters

Gradient descent update rule:
$W := W - \alpha * \partial J/\partial W$

5. Core Conceptual Lessons

- Matrix multiplication encodes linear transformations.
- Columns of a matrix represent transformed basis vectors.
- Model choice (linear vs sigmoid) changes gradients and learning behavior.
- Determinant measures area scaling.
- Vectorization removes loops and enables efficient training.